AssetLotse

MANUAL

AssetLotse

Issue equipment, take it back and keep track of it – for staff, managers and administrators.



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AGPL-3.0-or-later
helfi.space

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PART A

For everyone

What everyone who receives equipment from AssetLotse needs to know: signing in, keeping track of your own items, extending, returning and – if enabled – looking up the BitLocker key of your own device.

1 AssetLotse at a glance

AssetLotse manages your institute's equipment that is issued and returned: laptops, USB sticks, external hard drives, headsets, projectors, keys, work phones, cables and the like. Consumables do not belong in it. The program runs in the browser – there is nothing to install.

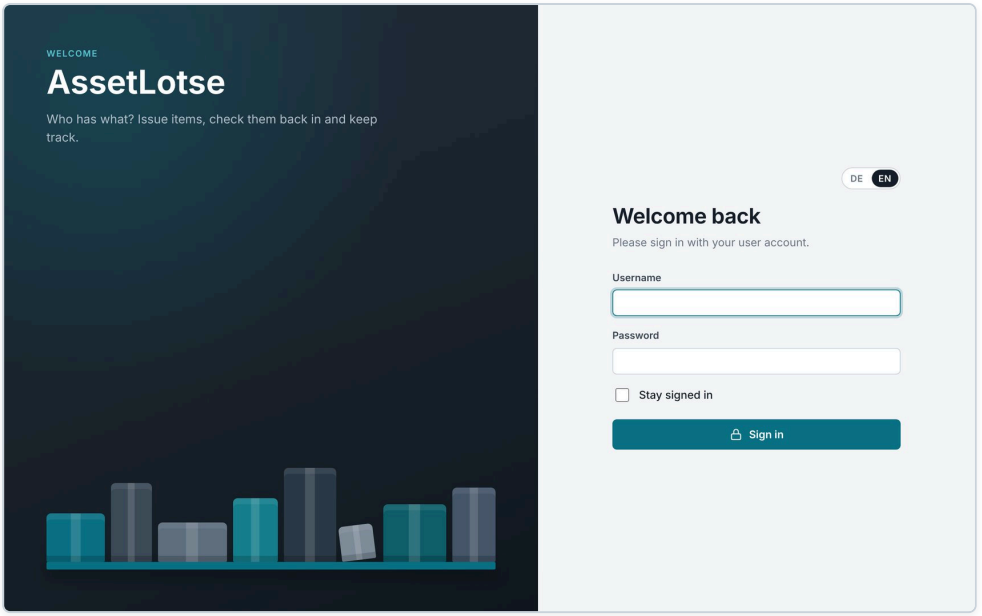
Every person has one of three **roles**:

ROLE	CAN
Staff member	see their own items, extend loans with a return date (if allowed), set notifications and language. Depending on the settings, also view the inventory and check out or return items themselves.
Manager	add, edit and delete items, issue and check them in, use the issue desk, print labels, export the inventory – for the items of their own groups and all "general" items.
Administrator	everything, plus users, groups, settings, Active Directory, email, branding and backup.

Groups are, for example, departments or teams. Every item either belongs to a group or is **general**. General items can be issued to anyone, group items only to members of that group.

> **How this manual is organized:** Part A is for everyone, Part B for managers, Part C for administrators. It ends with chapters on troubleshooting, emergency commands and an appendix with all settings of the `.env` file. The program itself also has a **Help** section that adapts to your settings.

2 Signing in, language and help



The sign-in page (your administrator can customize the logo, colors and texts).

Open the address your IT team gave you (for example `https://assets.institut.de`). Sign in with your **username and password**. If AssetLotse is connected to Active Directory, this is your normal Windows account – the same name and password as on your PC. With **Stay signed in** you won't have to sign in again on this device for a while.

After several wrong passwords, sign-in is locked for a few minutes. Wait a moment or ask an administrator to lift the lock.

Two-factor authentication

If you have set up two-factor authentication (or it is mandatory for you), AssetLotse asks, after your password, for a code from your authenticator app or for your security key. You can set this up under **Security** in the sidebar – see the chapter Profile and security.

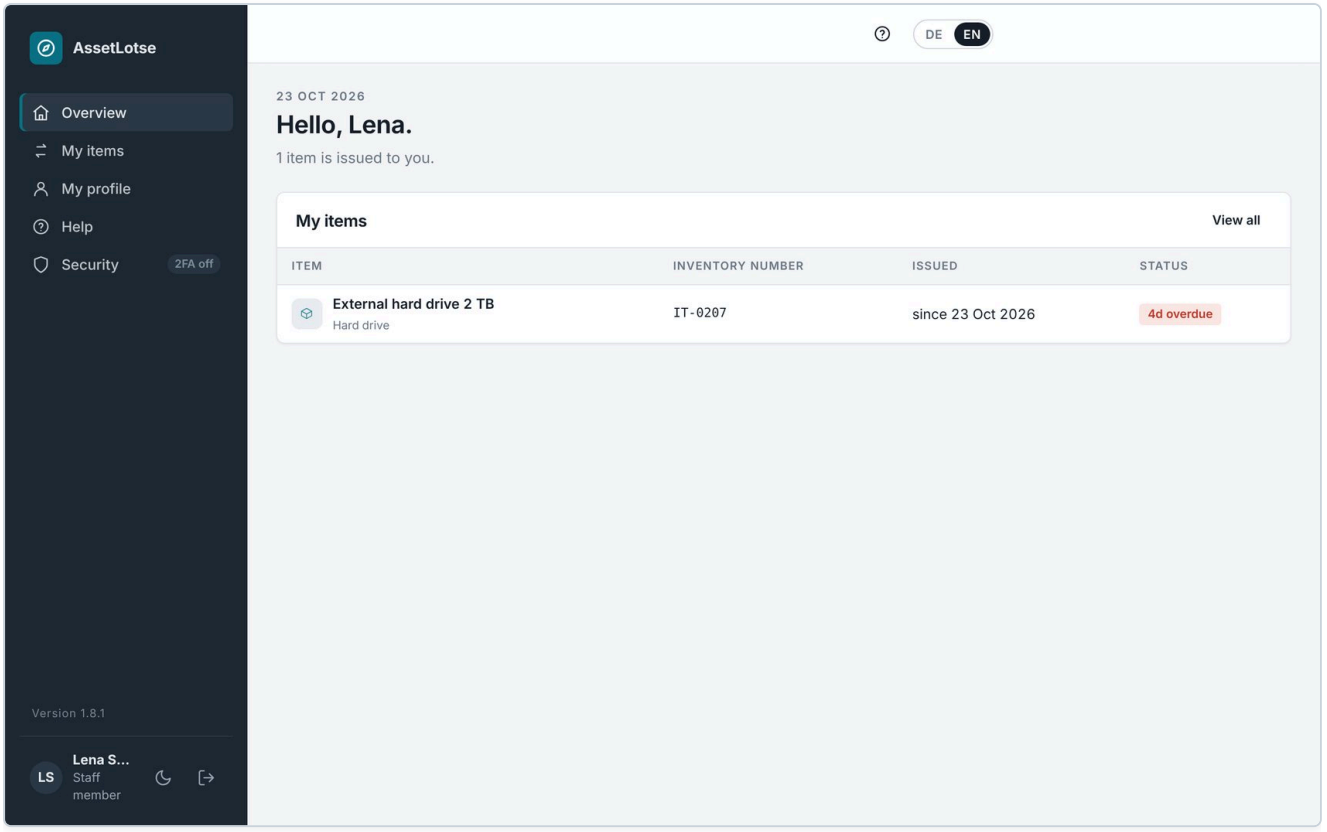
Language

At the top right, **DE** / **EN** switches between German and English. Your choice is saved in your profile and also applies to emails sent to you.

Help in the program

Under **Help** you will find explanations of all pages you are allowed to use. The **?** at the top right jumps straight to the help for the page you are on. The help texts adapt to your settings – for example, they state the actual loan period.

3 My items



A staff member's overview: what is currently issued to her, with status.

The **Overview** and the **My items** page show everything currently issued to you – with inventory number, issue date and status. The **Overdue** and **History** tabs show only overdue items and items already returned, respectively.

STATUS	MEANING
open-ended	You keep the item until you return it (the default, e.g. for a work laptop). It never becomes overdue, and there are no reminders.
until 14 Nov 2026	A return date has been agreed, and there is still plenty of time.
due 25 Oct 2026	The return date is close – return or extend soon.
4d overdue	The return date has passed.

Clicking an item opens its **detail page** with inventory and serial number, location and accessories.

> **No "Inventory" in the menu?** That is normal: by default, staff members only see their own items. Items are issued by the management team or at the issue desk. If your administrator makes the inventory visible, the **Inventory** menu item appears, with a search.

Extending

If an item has a return date and self-service extension is allowed, **Extend** moves the date (default: by 14 days, at most twice per loan). It counts from the later of today and the current return date. Once the maximum is reached, the button disappears – then only the management team can extend. Open-ended loans need no extension.

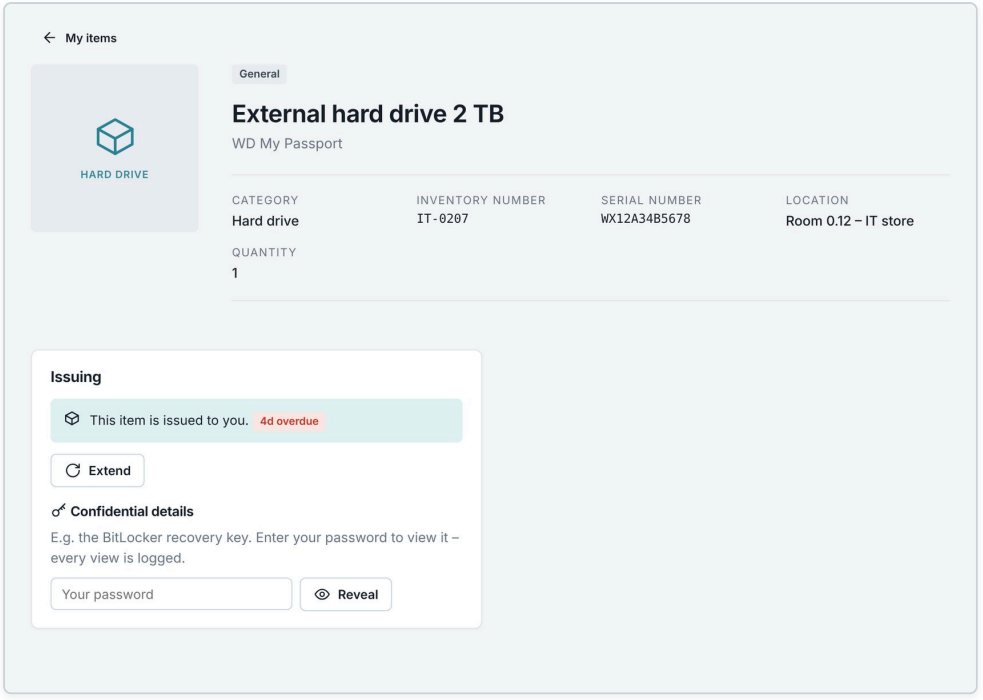
Returning

Bring the item with all accessories to the management team or the issue desk; the return is recorded there. Only then does it disappear from your items. If self-return is enabled, there is also a **Return** button in the list (**Return** on the item page) – afterwards, bring the item to its location.

Email reminders

For items with a return date, AssetLotse can send a reminder shortly before and an overdue notice once it is overdue. Several items are combined in one email. You can turn this off in your **Profile**.

4 BitLocker key of your own device



On the detail page of the issued hard drive: revealing the key after entering your password.

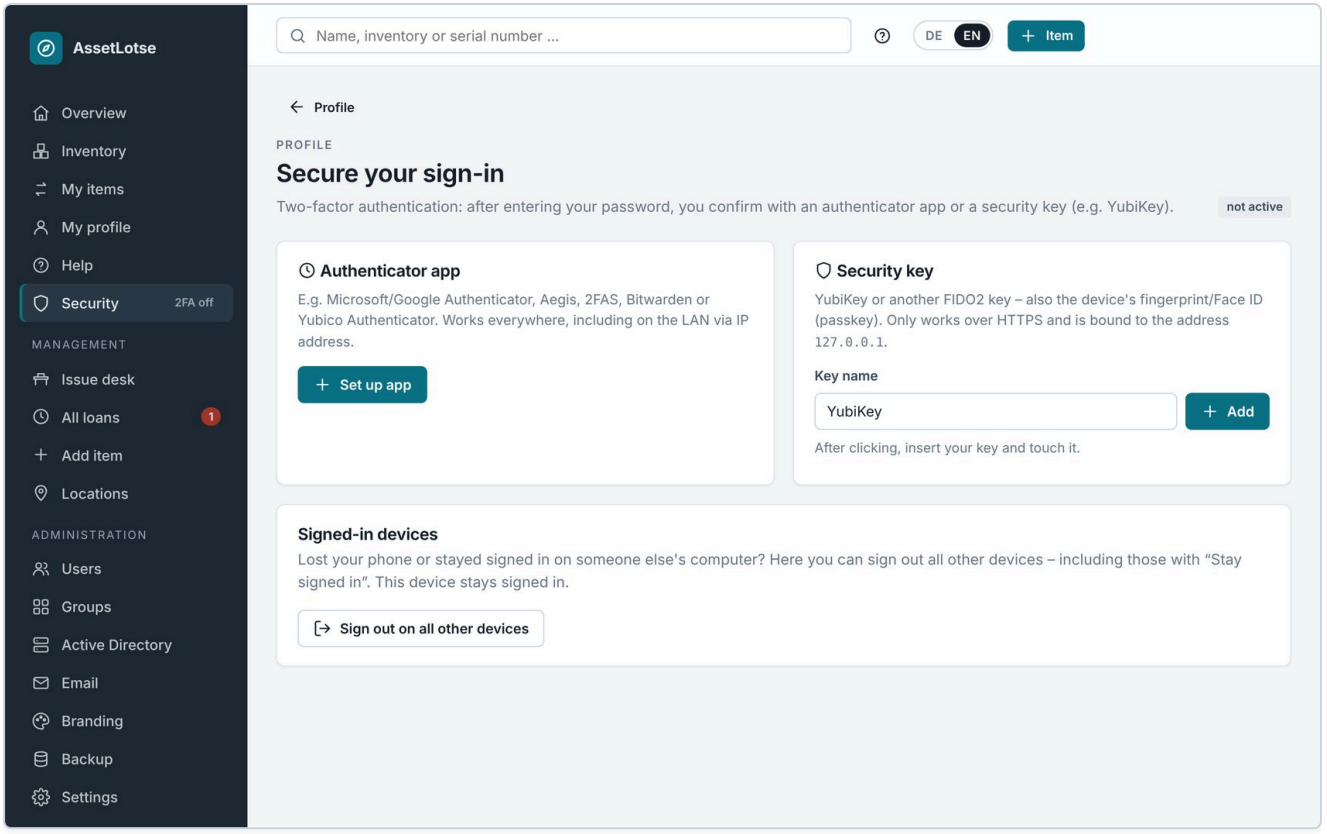
For encrypted devices (e.g. an external hard drive with BitLocker), the management team can store the **recovery key** in AssetLotse. If they have enabled it for your device, you can look it up yourself if you have forgotten it:

1. Open the device under **My items**.
2. Under **Confidential details**, enter your password and click **Reveal**.
3. The key appears with a **Copy** button. **Hide** hides it again.

This only works while the device is issued to you – once it is returned, access ends. Every reveal is logged. Other staff members never see your key. If you have set up two-factor authentication but signed in without the second factor (e.g. on the internal network), you must confirm it first.

5 Profile and security

Under **My profile** you set your **language**. If sending emails is set up, you also enter your **email address** there (for Active Directory accounts it comes from AD automatically) and turn **email notifications** on or off. Local accounts can **change their password** here (at least 10 characters). On the left you see your loan history.



Security: authenticator app, security keys and signed-in devices.

Under **Security** you set up **two-factor authentication**:

- **Authenticator app** (Microsoft or Google Authenticator, Aegis, 2FAS, Bitwarden …): scan the QR code, enter the code shown, done.
- **Security key** (YubiKey, FIDO2) or your device's fingerprint/Face ID (passkey): only works over HTTPS. It's best to register a spare key right away.
- **Recovery codes:** when you set things up, ten one-time codes appear – only once. Print them or keep them somewhere safe; they let you in even without your phone or key.

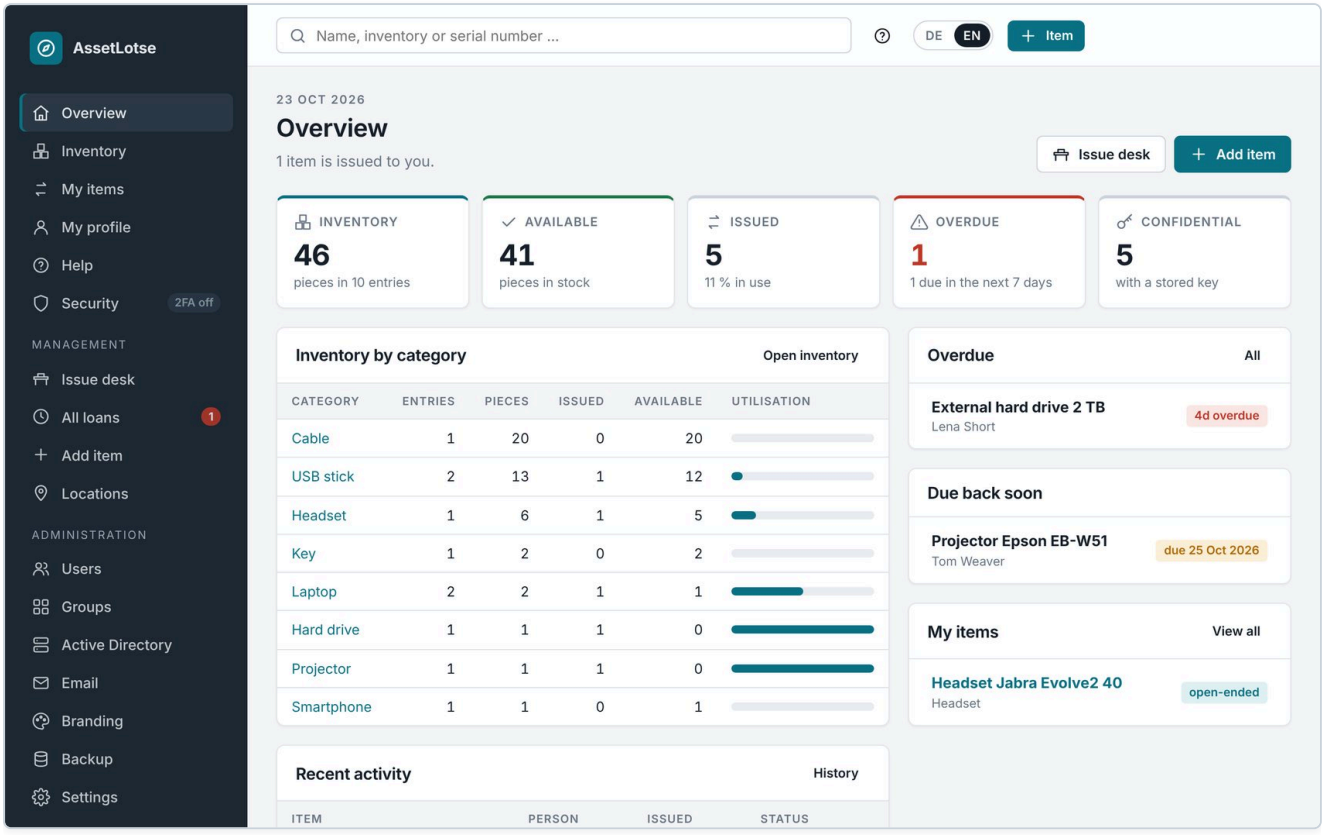
Sign out on all other devices ends all other sessions – useful if you have, for example, signed in on someone else's computer.

PART B

For managers

Adding, issuing and checking in items – at the issue desk often with a single scan. Managers work with the items of their groups and all general items; a manager without a group manages the entire inventory.

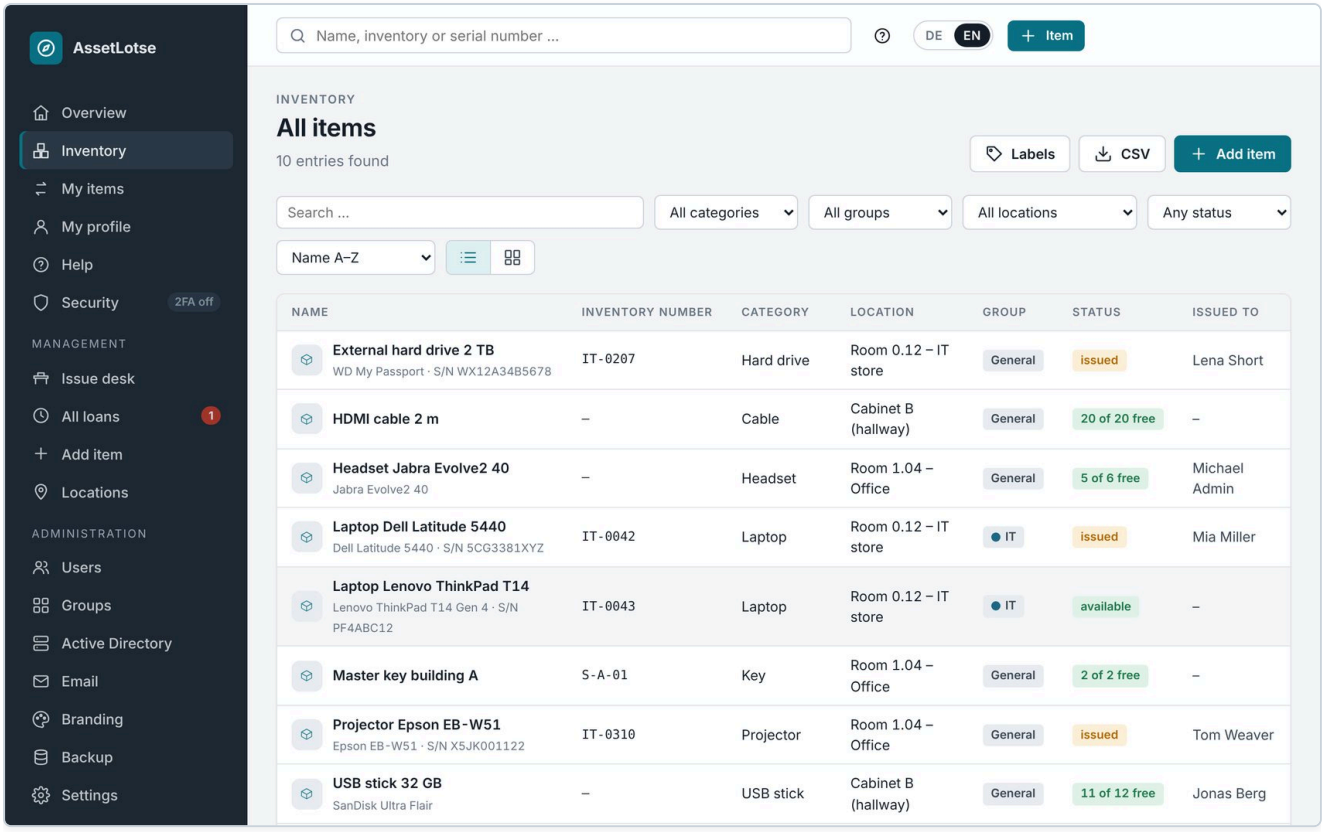
6 Overview with key figures



The start page for managers and administrators.

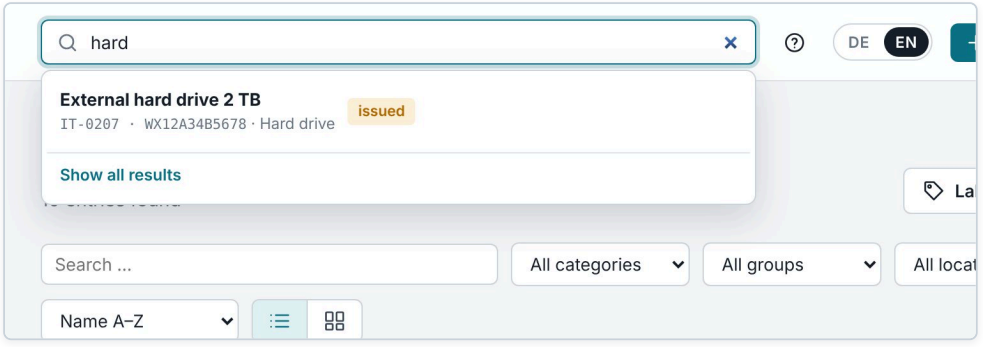
The key figures are at the top: **Inventory** (pieces and entries), **Available**, **Issued** with utilization, **Overdue** (with those due in the next 7 days) and **Confidential** (items with a stored key). Clicking one of the first four tiles opens the matching filtered list. Below: **Inventory by category** with utilization bars, **Recent activity**, **Overdue** and what is **Due back soon**.

7 Inventory: search, filter, view



The inventory as a table with filters, labels and CSV export.

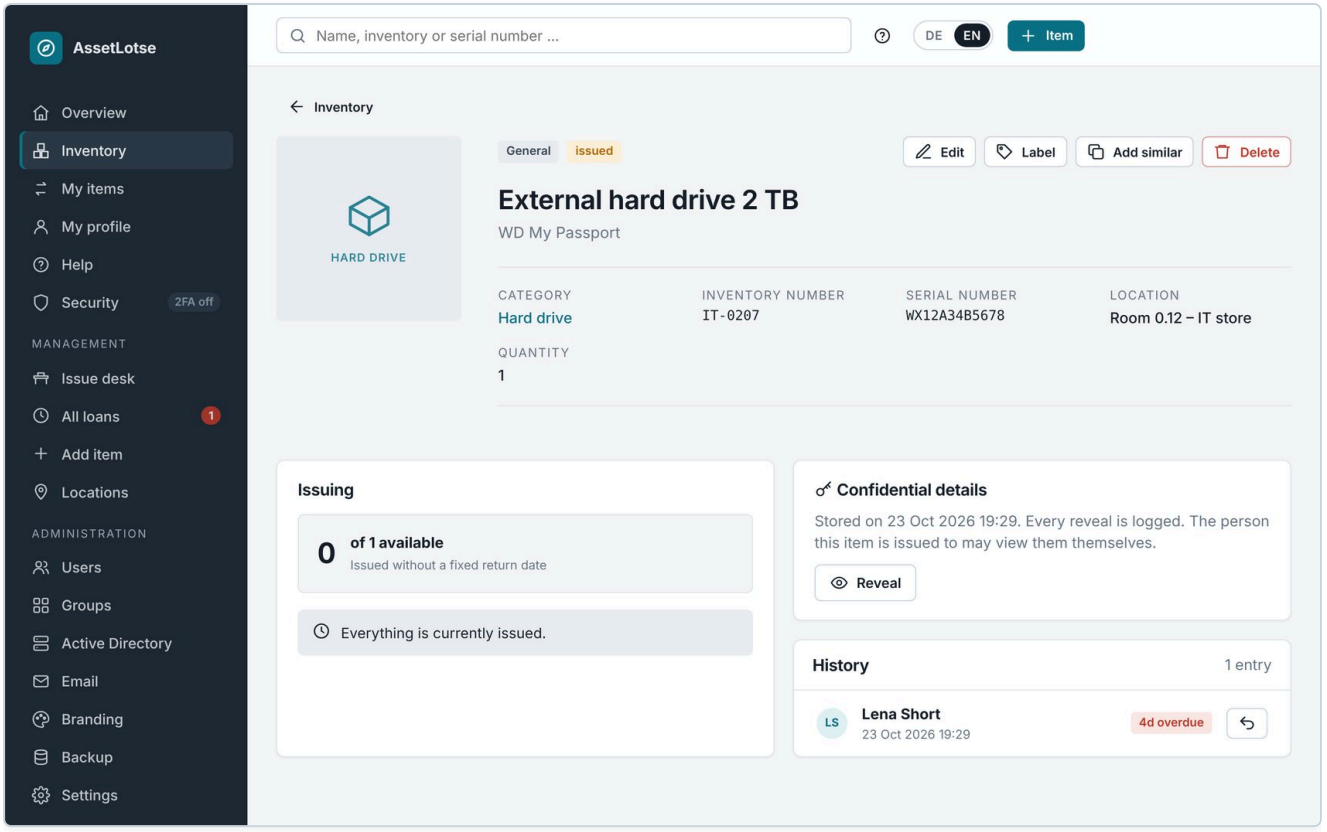
The **Inventory** shows the items of your groups and all general items (all items with the “whole inventory” setting) – by default as a table with the **Issued to** column; on the right you can switch to tiles with photos. You can filter by **category, group, location** (once locations have been defined) and **status** (available/issued), and sort by name, category, inventory number or date added.



The search at the top shows matching items as you type.

The **search field at the top** shows matching items as you type, with inventory and serial number and whether they are available. A click (or `↓` and `Enter`) opens the device; `Enter` on its own or **Show all results** opens the full list. The suggestions search name, category, manufacturer, model, inventory and serial number; the full list also searches location and description. You can also scan an inventory number from the label with a barcode scanner.

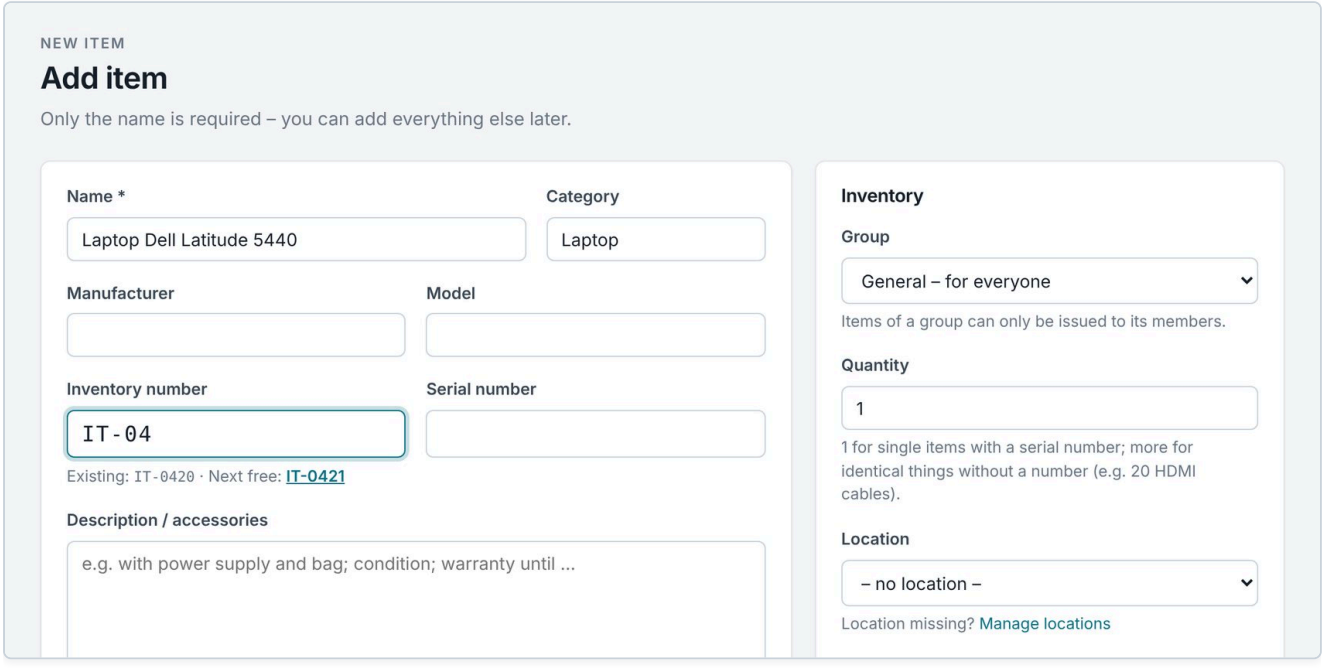
The detail page



Detail page of an issued hard drive with confidential details and history.

It shows all details, the current status and the history. At the top you will find **Edit**, **Label**, **Add similar** and **Delete** (only possible if the item is not issued). At the bottom left you issue the item under **Issue to …**; on the right are the confidential details and the history of all loans.

8 Adding and editing items



As you type the inventory number, AssetLotse shows existing numbers and the next free one.

Click **+ Item** at the top. Only the **name** is required (e.g. “Laptop Dell Latitude 5440”). You can add everything else later:

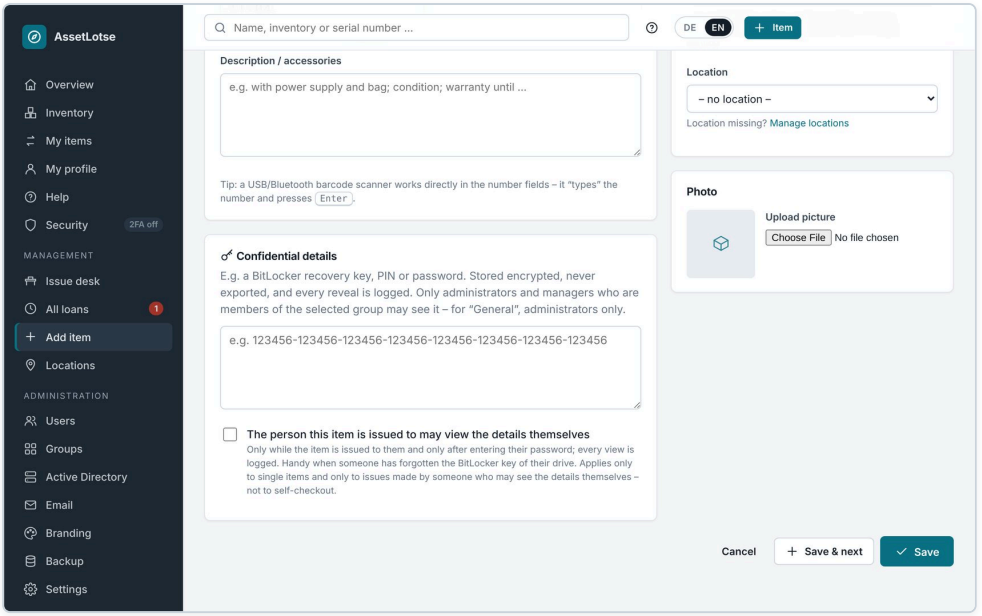
FIELD	NOTE
Category	free text, e.g. “Laptop”, “USB stick”, “Key”. Existing categories are suggested; you create a new one simply by typing it.
Manufacturer, Model	free text
Inventory number	e.g. from the label, also via barcode scanner. As you type, the field shows **Available** or **Already in use** (with a link to the device), plus similar existing numbers and the **Next free:** number – one click takes it over (e.g. `IT-0044` after `IT-0043`). Inventory numbers must not start with `#` (that is reserved for the internal number).
Serial number	also checked for “available/already in use”.
Description / accessories	e.g. “with power supply and bag; warranty until 2028”
Group	**General – for everyone** or a group; group items can only be issued to its members.
Quantity	`1` for single items with their own number. Use more only for identical things without a number, e.g. 20 HDMI cables – they are then issued piece by piece.
Location	from the list of locations (see Locations).
Photo	JPG, PNG, WEBP or GIF; it is scaled down and saved without metadata.

> **Each number exists only once.** An inventory or serial number that is already used by another item is rejected when saving – regardless of upper or lower case. Your input stays in the form. This way no duplicates can arise from typos or copying.

Adding many identical devices

- **Save & next** immediately opens a new form with the same details after saving – without numbers, photo and confidential details. For ten identical laptops you then only need to scan the inventory and serial numbers.
- **Add similar** on the detail page does the same, starting from an existing item.

9 Confidential details (e.g. BitLocker keys)



The "Confidential details" box in the form.

In the **Confidential details** box you store, for example, the **BitLocker recovery key**, a PIN or a password. These details

- are stored **encrypted** and never exported or sent by email,
- are visible only to **administrators** and to **managers who are members of the item's group** – for "General" items only to administrators. Tip: assign devices with a key to a group such as "IT".
- only appear after clicking **Reveal**; every reveal, change and deletion is recorded with name and time in the **Log** on the detail page.

When editing, the field stays empty – only fill it in to **replace** the details. Tick **Delete confidential details** to remove them. Only people who may see the details themselves can delete items with confidential details or assign them to another group.

Access for the person who has the device

With the box **The person this item is issued to may view the details themselves**, someone can, for example, look up the BitLocker key of their hard drive themselves (see Chapter 4). So that nobody can "borrow" their way to a key, this applies

- only to **single items** (quantity 1),
- only if the device was issued by someone who **may see the details themselves** (an administrator or a manager of the group) – not with self-checkout,
- only after re-entering your own password, and only while the device is issued.

10 Issuing and checking in

You handle individual loans directly on the **detail page** under **Issue to …**: choose the person (only people who may receive the item are listed), then either **Return by** a date or **open-ended**, and optionally a note such as "with power supply". If the default loan period is 0, **open-ended** is preselected

– to set a return date, untick it first. You can check items in from the history on the same page or under **All loans**.

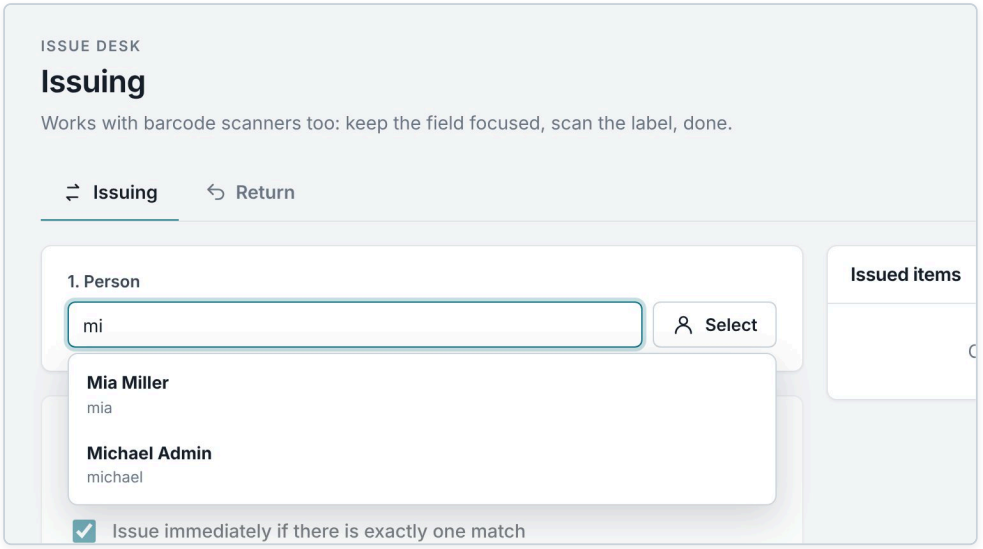
AssetLotse automatically checks whether the account is active, the person may receive the item (group), a piece is still available, they don't already have it and – if configured – whether they have reached their maximum number of items. If sending emails is set up, the person receives a confirmation.

For many loans in a row – for example when new colleagues receive their equipment – the **issue desk** is faster.

11 The issue desk

Issue desk in the sidebar has two tabs: **Issuing** and **Return**. It is designed for the keyboard and barcode scanners – the input field stays active, so you simply scan one label after another.

1. Choose the person



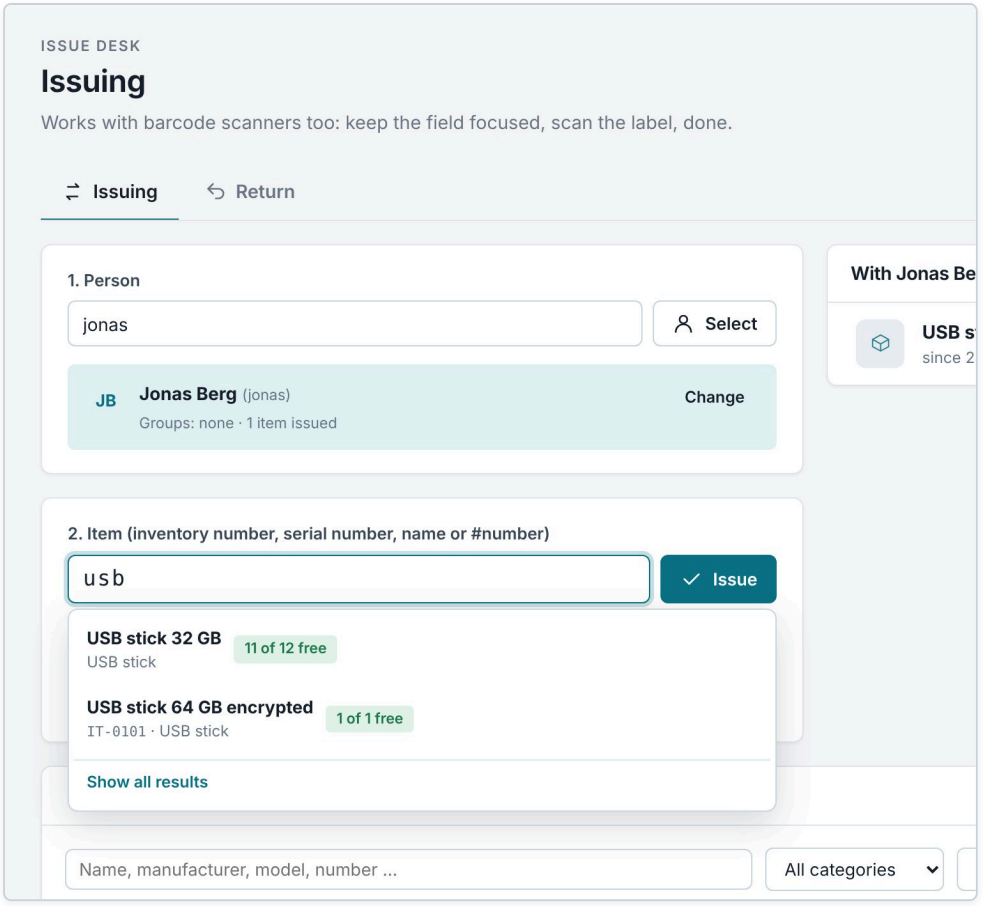
Accounts and people from Active Directory appear as you type.

Type a username or name. Matching accounts appear as you type; a click selects the person. If Active Directory is set up, people who have **never signed in** also appear – marked **AD – no account yet**. A click creates their account (AssetLotse checks the person directly in AD once more) and selects them. Only people who would be allowed to sign in under your AD rules are found.

On the right you see what the person already has (of the items you manage). **Change** starts over with the next person.

At the desk the **default loan period** always applies; you set a different return date on the item's detail page.

2. Scan or search for the item



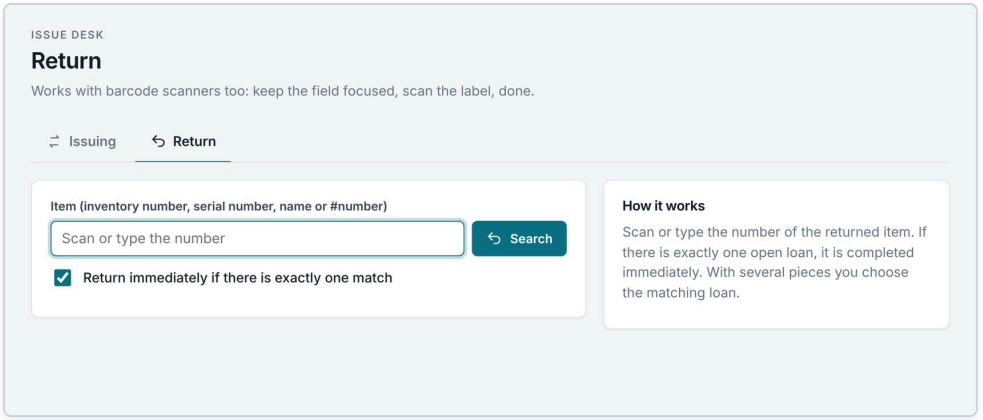
As you type, items the selected person may receive appear – with “x of y free”.

Scan the label or type the inventory number, serial number or name. As you type (e.g. “hard”), matching items that this person may receive appear; a click (or `↓` and `Enter`) issues the item immediately. Items the person may not receive or that are out of stock are marked accordingly.

- **Issue immediately if there is exactly one match** (on by default): if exactly one item matches, it is issued without another click, and the field is ready for the next one right away. This happens automatically only for exact matches (inventory number, serial number, `#number`), never when searching by name.
- `#12` always stands for **internal number** 12 – that is how it is encoded on labels of items without an inventory number.
- **Search the inventory** (box at the bottom of the page): search text and/or category, by default only available items you may issue – with **Issue** directly in the results list.

> **Barcode scanner:** a USB or Bluetooth scanner “types” the number and presses Enter – this works without any setup. The suggestion list does not get in the way: Enter without a selected suggestion always searches for the scanned number. If no Enter arrives, configure the scanner for “Enter/CR suffix” according to its manual. Wrong characters (e.g. `ß` instead of `-`) mean that the scanner’s keyboard layout does not match the computer’s.

Return



The Return tab.

In the **Return** tab you scan the label. With **Return immediately if there is exactly one match**, the loan is closed right away if there is exactly one open loan. For items with several pieces (e.g. cables), choose the right loan and click **Check in**. Take a quick look at the accessories listed in the note.

12 Printing labels

For every item, AssetLotse generates a **PDF label in the 62 × 29 mm format** – suitable, for example, for Brother QL printers with DK-11209 labels or 62 mm continuous tape. It shows the name, category/model, serial number, a **barcode** with the inventory number and the name of your organization.

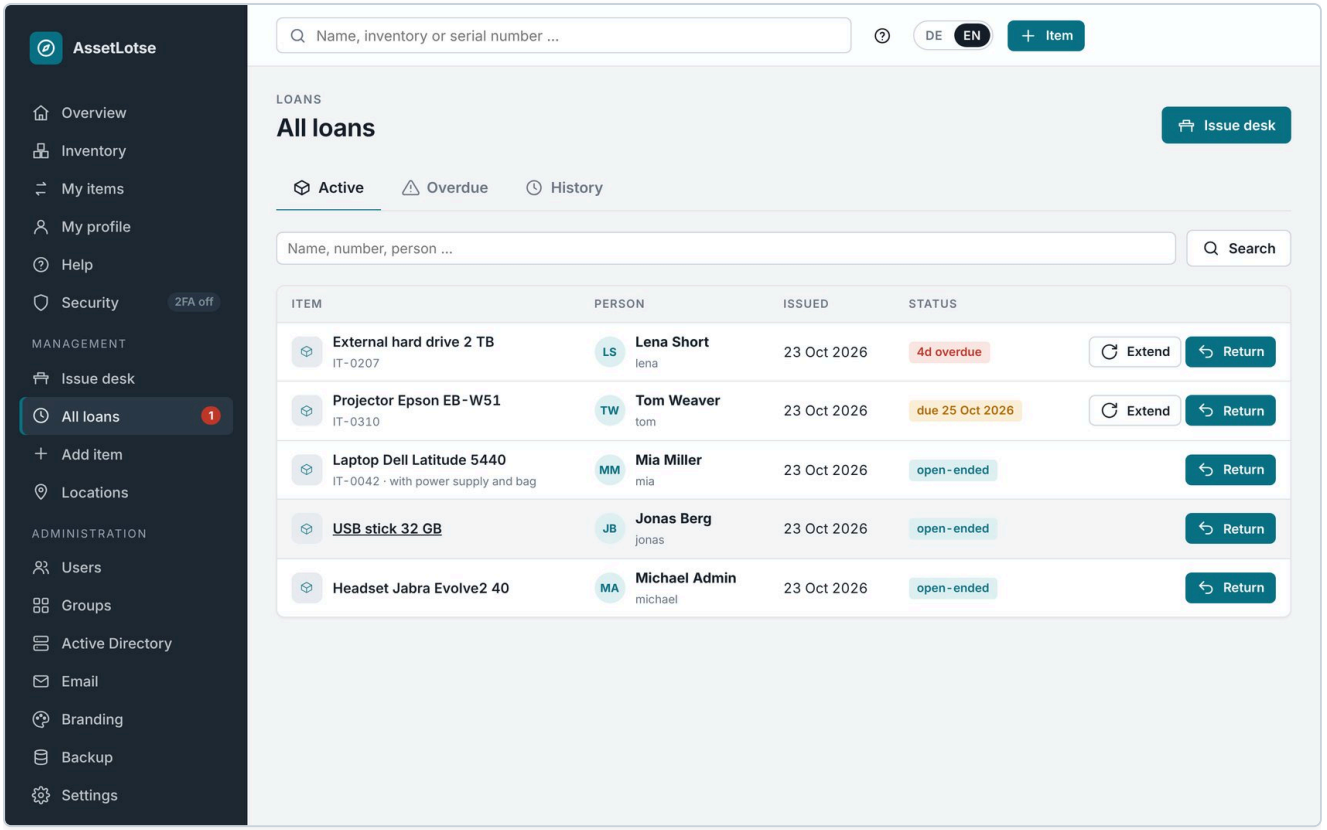


62 × 29 mm label with a Code 128 barcode.

- **Single label:** detail page → **Label**.
- **Many at once:** filter the inventory (e.g. category “Laptop”) → **Labels**; one PDF with one page per item (at most 500).
- Without an inventory number – or if it contains umlauts or other special characters, or is very long – the barcode encodes `#<internal number>`. The issue desk finds both.

Printing: in the print dialog, choose the 62 × 29 mm label size and set **“Actual size” or 100 %** – not “Fit to page”. For continuous tape, set the length to 29 mm in the Brother driver. For heavily used devices, plastic labels (e.g. Brother DK-22113) last much longer than paper.

13 All loans, reminders and export



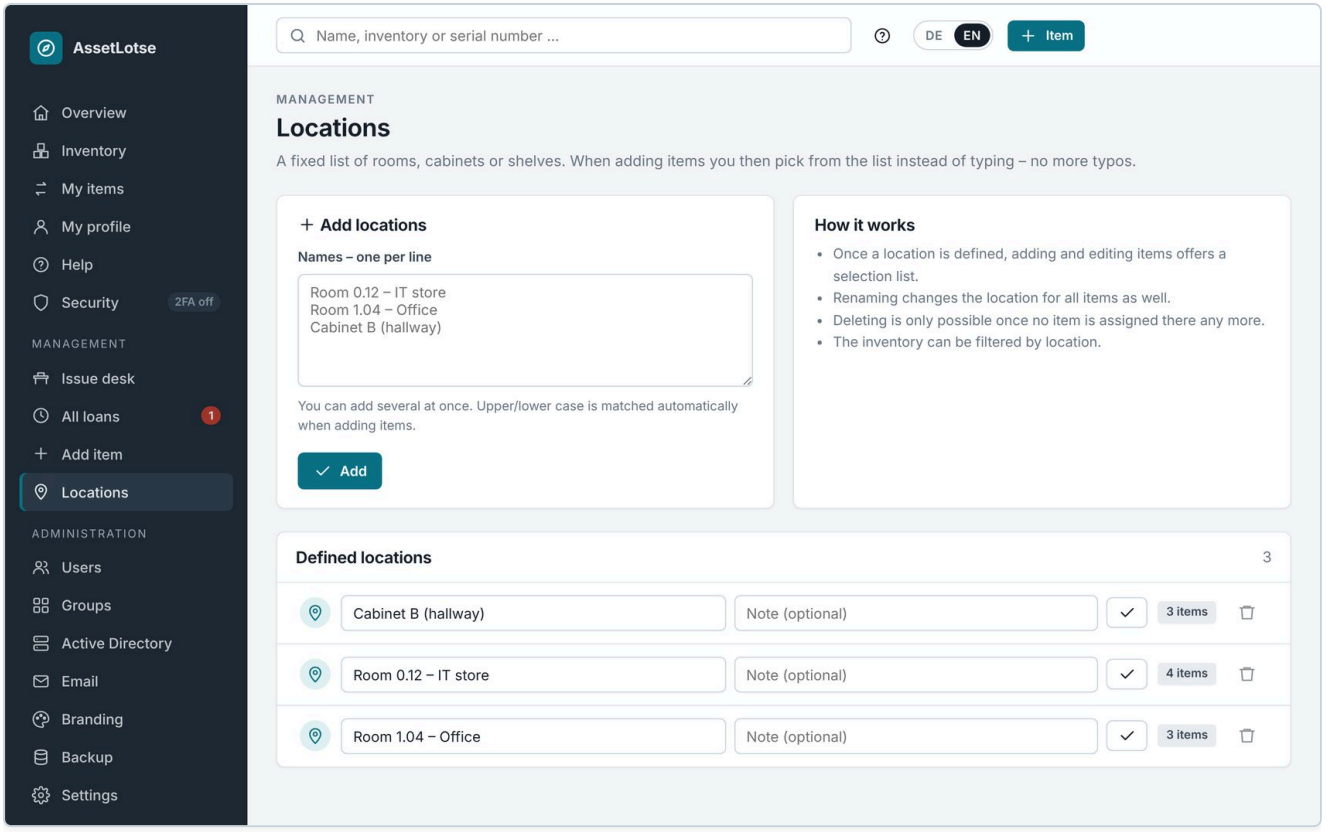
All loans with Extend, reminder and Return.

All loans shows the active loans of your groups, the **Overdue** tab only overdue ones, and **History** the completed ones. On each row you can **Extend**, send an **email reminder** with the envelope icon (for loans with a return date, if sending emails is set up), and check the item in with **Return**. Managers can always extend, even beyond the maximum.

CSV export

In the **Inventory**, **CSV** downloads all items you may manage as a spreadsheet – regardless of any filters set (semicolon, UTF-8 – opens directly in Excel): ID, name, category, manufacturer, model, serial number, inventory number, location, group, quantity, available and **issued to**. Confidential details are never exported.

14 Locations



Defining locations – when adding items, you then pick from a list instead of typing.

Under **Locations** you define rooms, cabinets or shelves once – several at a time if you like, one per line. After that, the location is **picked from a list** when adding and editing items; this prevents spelling variants.

- **Renaming** changes the location for all affected items as well. **Deleting** is only possible once no item is left at that location.
- Locations that were typed freely in the past appear under **In the inventory, but not defined**. You can add them as locations or replace them with a defined location – if the spelling is similar, the matching one is already preselected.
- Renaming and replacing that affects items of other groups is reserved for administrators and managers without a group.

PART C

For administrators

Installation, setup and operation: users and groups, settings, Active Directory, email, two-factor rules, backup, security and privacy.

15 Installation and updates

AssetLotse runs as a **Docker container** on a Linux server. You need Docker with Docker Compose (the `docker compose` plugin or the older `docker-compose` – any version) and `unzip`.

First installation

1. Copy the package to the server and unpack it, e.g. to `/opt/assetlotse`.
2. In that folder, run: `docker compose up -d --build`
3. Open `http://SERVER:9343` in the browser and create the **local administrator account**. This account always works – even if Active Directory is unavailable. Setup in the browser only works from the internal network; otherwise, create the account on the server with `create-admin` (see Emergency commands).

All data (database, photos, keys, backups) is stored in the `./data` subfolder. The container does not run as root but as user `1000:1000` (can be changed with `PUID` / `PGID`).

Your own settings: the .env file

Port, time zone and so on belong in the `.env` file in the program folder – not in `docker-compose.yml`. The `.env` file is never overwritten by updates. The most important settings are listed with comments in `.env.example`, and all of them in the Appendix. Important to start with:

- **Different port:** add `ASSETLOTSE_PORT=8080` and run `docker compose up -d`. The value applies both outside and inside the container; 1024–65535 are allowed.
- `ASSETLOTSE_BIND=127.0.0.1` makes AssetLotse reachable **only on the server itself** – intended for a reverse proxy that runs directly on this machine (not as a container). If the proxy runs as its own container (e.g. Nginx Proxy Manager), it is better to put both into a shared Docker network. For access from the network, leave this line out.

Updating

You install a new package with the `update.sh` script:

```
./update.sh assetlotse-1.7.0.zip
```

The script checks the package (it shows the SHA-256 checksum and compares it if a `PACKAGE.zip.sha256` file is next to it; it refuses packages with symbolic links or foreign paths), creates a **backup**, only then replaces the program files (if anything goes wrong, the previous ones are restored), builds the image `assetlotse:VERSION` – if that fails, the old container keeps running unchanged – and restarts. `./data`, `.env` and `docker-compose.override.yml` are never touched. The database is migrated automatically on startup.

Options: `--no-backup`, `--no-cache`, `--prune` (also delete the previous image), `--no-cleanup` (do not clean up), `--cleanup` (clean up only, build nothing), `-y` (no questions, e.g. for cron).

Cleanup: after a successful update, the script cleans up by itself. The installed zip goes into the `packages/` archive in the program folder; the current and the previous package are kept there, older ones are deleted. Loose `assetlotse-*.zip` files in the program folder are moved to the archive, leftovers from aborted updates and older AssetLotse images are removed – the image of the previous version is kept for emergencies. Other Docker images on the server are not touched.

Going back to the previous version: install the previous package again with `./update.sh packages/assetlotse-OLD.zip`; if the new version has already changed data, also restore the backup taken before the update.

Ready-made images from GitHub

Instead of building on the server, you can let GitHub build the image. For every version tag in the repository (e.g. `v1.7.0`), GitHub Actions builds the image and publishes it as `ghcr.io/soleil0815/assetlotse`. The server then only downloads it:

```
docker login ghcr.io -u soleil0815     # once, password = token with read:packages
./update.sh --pull                     # newest version
./update.sh --pull --version 1.8.1     # pin a version, also for going back
```

Here too, the script backs up the database first, takes the installation files (`update.sh`, `docker-compose.yml` …) from the image and cleans up old images. You can still install a zip locally at any time – the image is then called `assetlotse:X.Y.Z`; the next `--pull` switches back. The script checks that the image comes from the correct GitHub repository and shows the version and digest.

Updates with other tools: after `./update.sh --pull`, the container follows the `latest` tag. You install new versions with `./update.sh --pull` or with a tool that updates containers (e.g. Dockhand or Watchtower). When a new version starts for the first time, AssetLotse itself creates a backup before any change to the database (`…-vor-update.db`). A tool does not replace `update.sh` and `docker-compose.yml` on the server; if they change, the CHANGELOG says so – then run `./update.sh --pull` once. `--version X.Y.Z` pins a version.

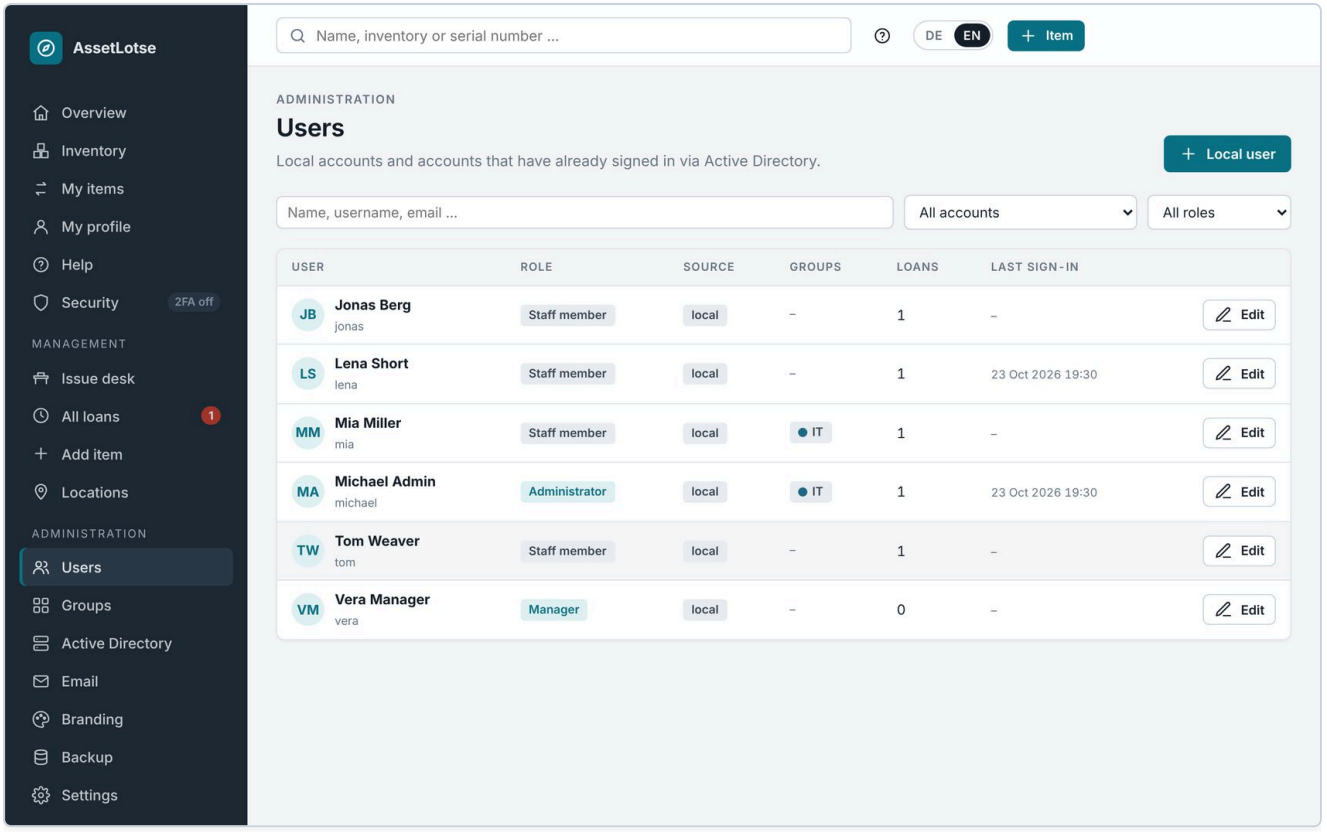
16 First steps after installation

A sensible order for the setup:

1. **Settings → General:** name of the organization (appears on labels and in emails), default language, inventory visibility.
2. Create **groups**, e.g. “IT”, “Accounting”, “Physics”.
3. Connect **Active Directory** (chapter Active Directory) – or create local users.
4. Set up **email** and send a test email.

5. Define **locations** (rooms, cabinets).
6. Set the **two-factor rules** – at least for managers and administrators if BitLocker keys are stored.
7. **Branding:** logo, colors, welcome text.
8. **Legal:** links to the legal notice and privacy policy.
9. Add the first **items** and print **labels**.
10. Check that the **backup** is running, and also copy the backups to another location.

17 Users, roles and groups



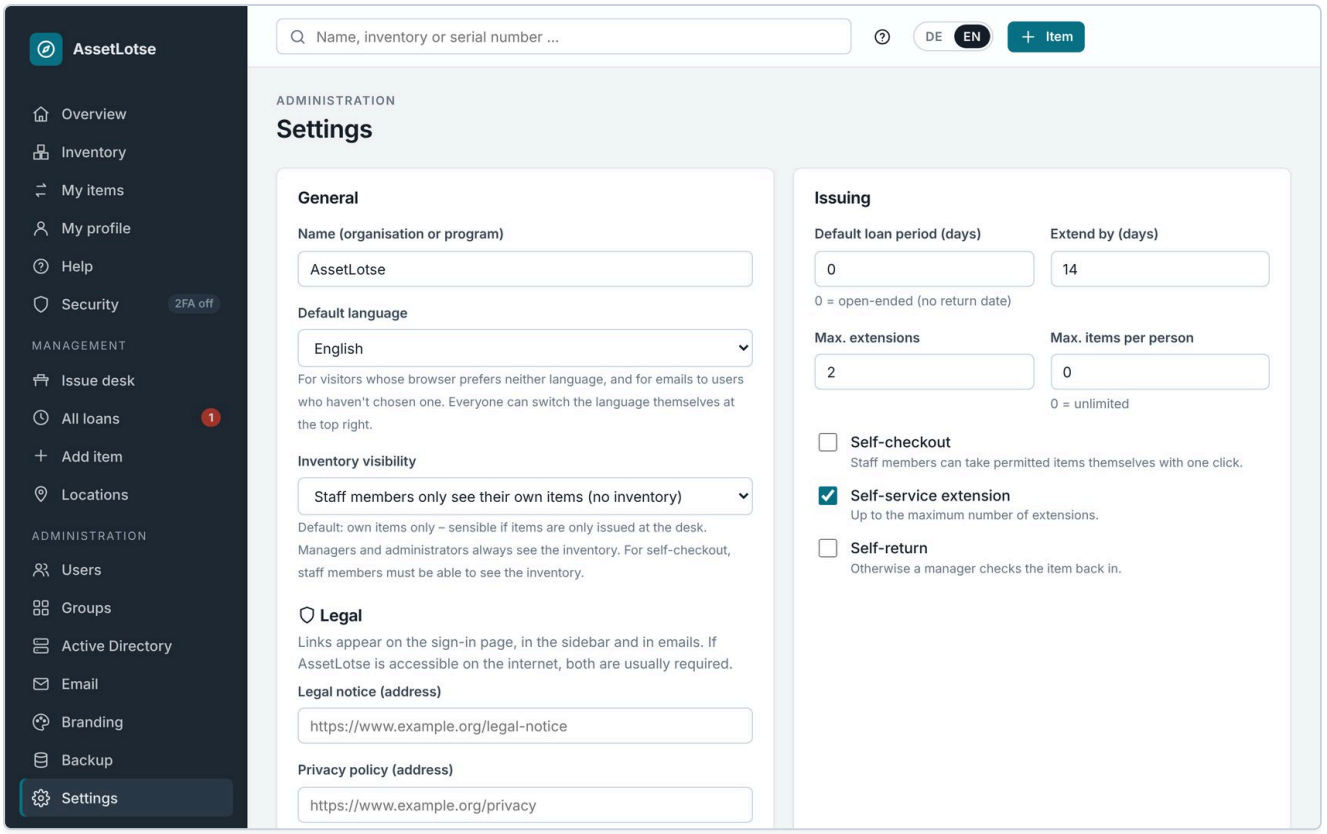
User management with filters by source, role and status.

Under **Users** you create **local accounts** (username, name, email, role, password with at least 10 characters) and edit all accounts. Accounts from Active Directory are created automatically at the first sign-in or at the issue desk (Take over); synchronization does not create accounts, it keeps existing ones up to date. The role from the AD rules applies from the person's first own sign-in – accounts taken over at the desk are staff members until then.

- You assign **groups** on the account – or automatically via a linked AD group (see below).
- Unticking **Account active** blocks sign-in; loans are kept.
- **Sign-in and security:** reset two-factor authentication (phone lost), **Lift lock** after too many failed attempts.
- **Deleting** is only possible if the person no longer has anything on loan; their loan history is removed.
- There must always be at least one active administrator; you cannot remove your own administrator rights.

Under **Groups** you create groups. If you enter the name of an **AD group** for a group (e.g. `Dept-Physics`) and **Assign groups automatically** is enabled, its members automatically become members of the AssetLotse group.

18 Settings

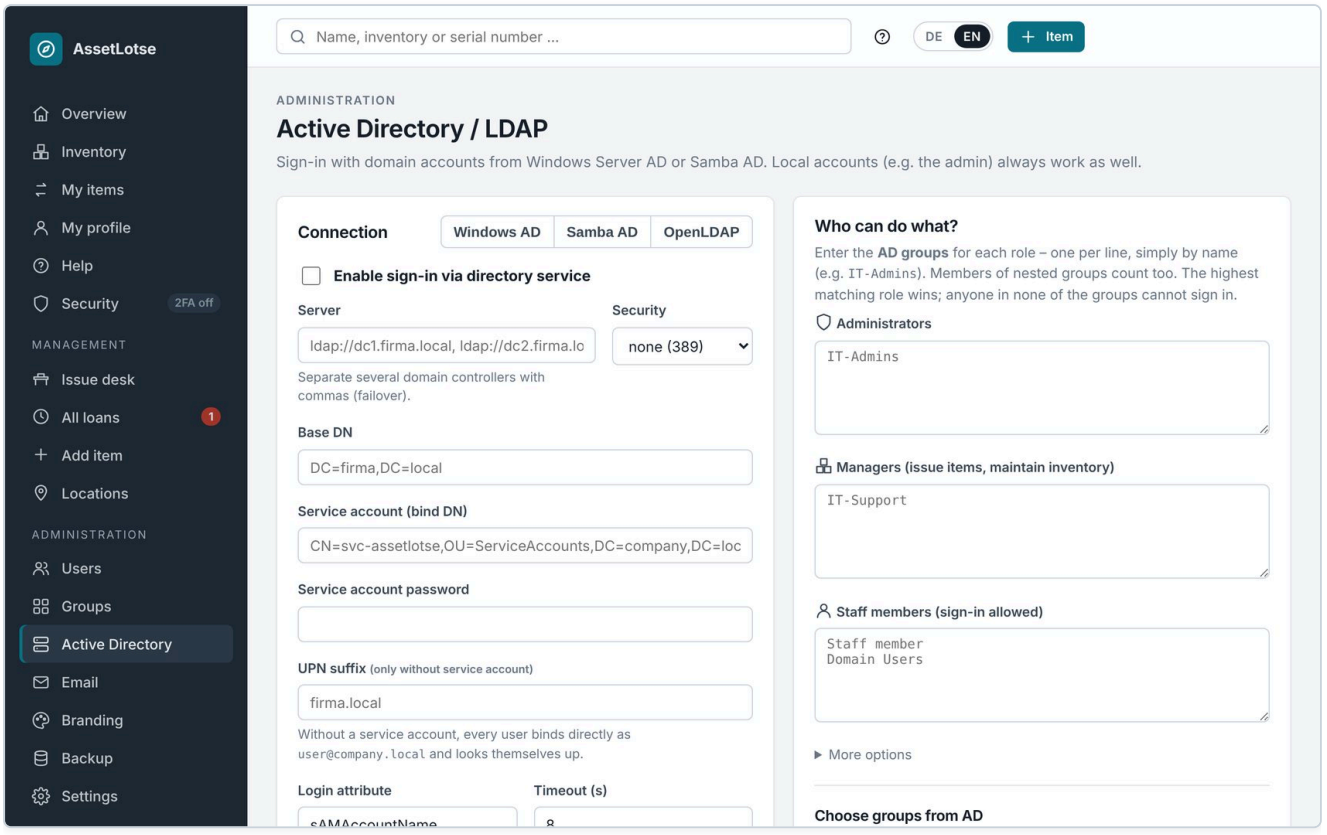


Administration → Settings.

SETTING	MEANING
Name	Name of the organization or the program – on labels, in emails and in the sidebar.
Default language	for visitors whose browser prefers neither German nor English, and for emails to users who have not chosen a language.
Inventory visibility	*own items only* (default – staff members see no inventory), *items of their own groups and general items* or *whole inventory*. Managers and administrators always see the inventory.
Default loan period	`0` = open-ended (default). With a number, every loan automatically gets this return date unless the manager chooses otherwise.
Extend by / Max. extensions	Default 14 days, at most 2 times.
Max. items per person	`0` = unlimited.
Self-checkout	Staff members take permitted items themselves (off by default; requires that they can see the inventory).

SETTING	MEANING
Self-service extension / Self-return	Default: extension on, return off.
Self-registration	off (default), with approval by an administrator, or account active immediately; optionally email confirmation, allowed domains and a group for new accounts.
Retention	how long completed loans are kept (default 12 months) – see Privacy.
Two-factor authentication	mandatory for nobody, administrators, managers and administrators, or all users; rules for the internal network – see Two-factor rules.
Legal	Addresses of the legal notice and privacy policy (sign-in page, sidebar, emails).

19 Active Directory



Administration → Active Directory.

Under **Active Directory** you connect AssetLotse to Windows AD, Samba AD or OpenLDAP – there is a template button for each.

1. **Server:** e.g. `ldaps://dc1.institut.local` (separate several DCs with commas). It's best to enter the DNS name as it appears in the certificate.
2. **Base DN:** e.g. `DC=institut,DC=local` – or, more narrowly, an OU.
3. **Service account** (read access only), e.g. `CN=svc-assetlotse,OU=ServiceAccounts,DC=institut,DC=local`. Without a service account, sign-in also

works directly via a UPN suffix – but then without the person search, without synchronization and without regular re-checking of sessions.

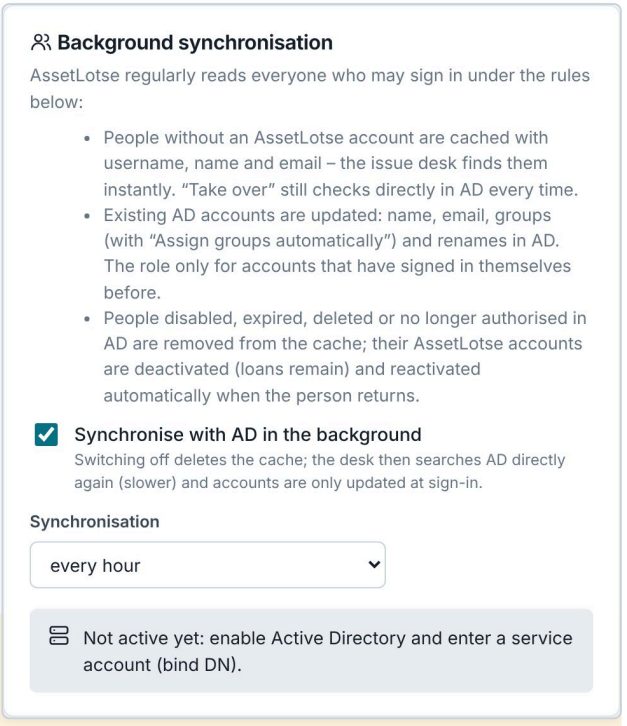
4. **TLS certificate:** **Fetch certificate from DC**, check the fingerprint, **Trust & save**. Where possible, AssetLotse fetches the root CA so that everything keeps working after the DC certificate is renewed.
5. **Who can do what?** For each role, enter the AD groups (or OUs), one per line – e.g. administrators `IT-Admins`, managers `AssetLotse-Managers`, staff members `Domain Users`. **Load groups** shows the groups from AD to click on. The highest matching role wins; anyone who matches no rule cannot sign in.
6. **Testing:** **Test connection** and **Test sign-in** show the DN, all groups and the role – even without saving first. Then click **Save**.

> Windows DCs and Samba usually refuse simple binds over unencrypted LDAP – use **LDAPS** (port 636) or **StartTLS**. Local accounts always take precedence; the local admin account works even if AD is down. If you are signed in with an AD account, you cannot save settings under which you would no longer be an administrator.

Background synchronization

With a service account, AssetLotse regularly synchronizes with AD (default: hourly; can be set to as little as once a day):

- **People without an account** are cached with username, name and email – the issue desk finds them immediately.
- **Existing accounts** stay up to date: name, email, groups, renames in AD (recognized by the fixed AD identifier). The role is only adjusted for accounts that have already signed in themselves.
- **Disabled, deleted or no longer authorized in AD:** the person is removed from the cache, their account is deactivated (loans remain visible) and automatically reactivated if they come back.

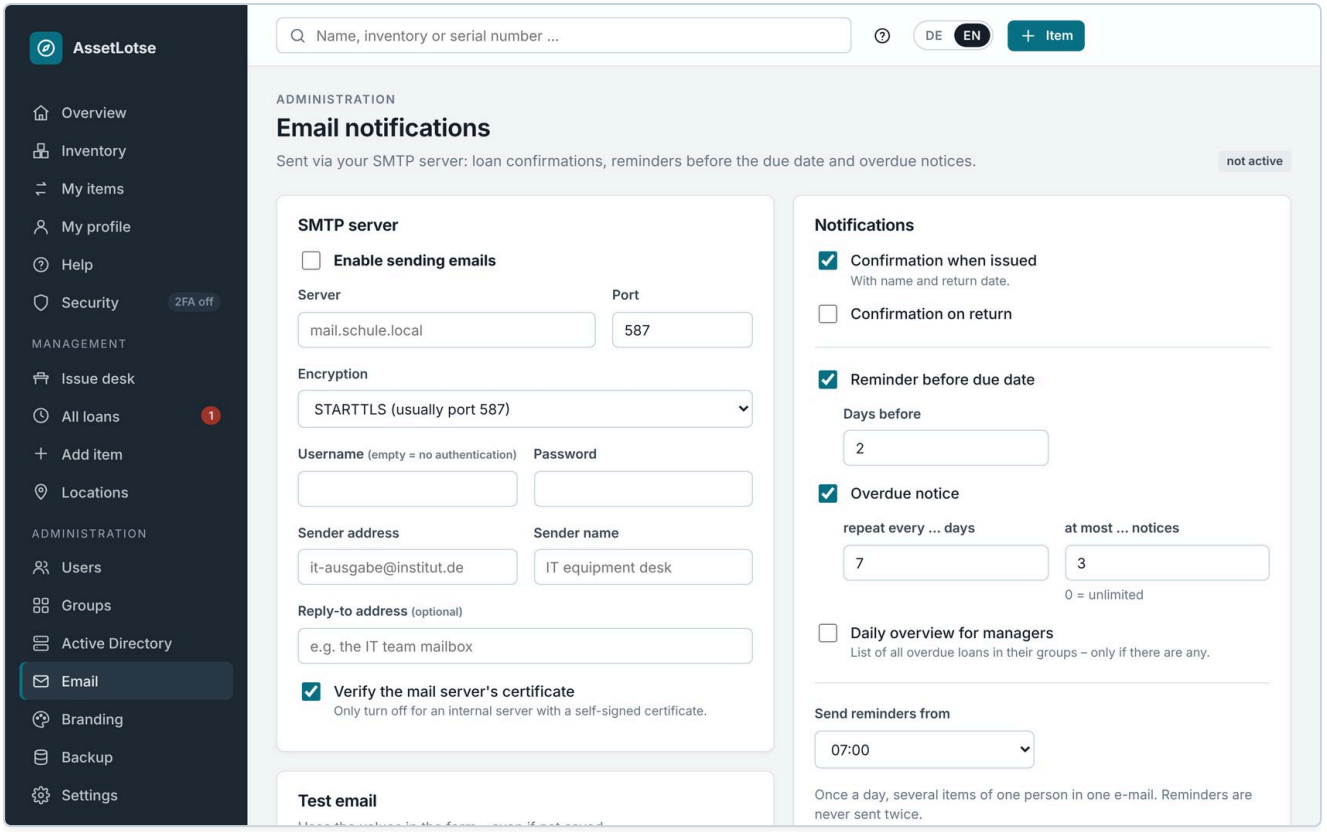


The "Background synchronisation" section with status and "Synchronise now".

> **Confirm once:** on the first synchronization – and whenever an unusually large number of accounts would be affected (more than 20 %), all accounts, an administrator or the last administrator – AssetLotse does not deactivate anyone automatically. The AD page then lists the affected usernames; check them and confirm with **Deactivate these accounts**. If a setting is wrong (base DN, filter, rules), correct it first.

If a username in AD is reassigned to a **different person**, that person never gets the old account with its history. In that case, rename the old account under **Users** (e.g. to "mia-old"). Older locked accounts without an AD identifier are never reactivated automatically – that is your decision.

20 Email notifications



Administration → Email.

Enter the SMTP server, port, encryption, credentials and sender, then click **Send test email**. The **Address of AssetLotse** (e.g. `https://assets.institut.de`) is needed for links in the emails.

NOTIFICATION	WHEN
Confirmation when issued	immediately, with the name and return date or “open-ended”
Confirmation on return	immediately (off by default)
Reminder	a few days before the due date – only for loans with a return date
Overdue notice	from the first day overdue, with configurable repetition (default: at most 3 notices)
Daily overview for managers	daily, only if there is something overdue in their groups (off by default)

Reminders and overdue notices run once a day from the configured time; several items of one person are combined in one email. Open-ended loans never get reminders. Typical servers: Microsoft 365 `smtp.office365.com:587 STARTTLS`, internal mail server `mail.institut.local:25`.

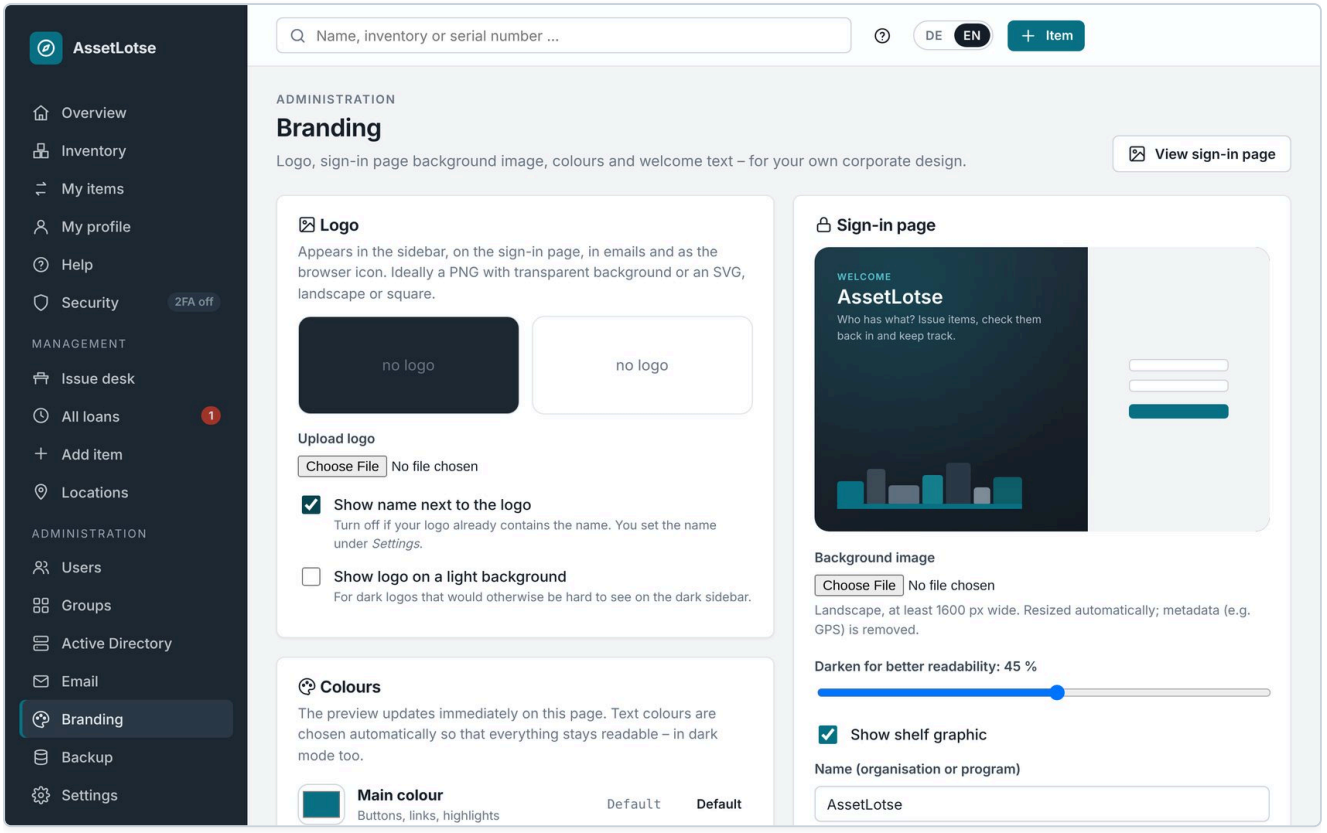
21 Two-factor rules and the internal network

Under **Settings** you decide for whom two-factor authentication is mandatory for **access from outside**: nobody (default), administrators, managers and administrators, or all users. Those affected are guided through the setup after signing in. If you store BitLocker keys, you should require it at least for managers and administrators.

For the **internal network** you can be more relaxed: *same as from outside*, *not mandatory* (users who have 2FA are still asked) or *no two-factor prompt*. You define what counts as “internal” by entering networks (e.g. `192.168.10.0/24`); the page shows how your current connection is classified. A session started on the LAN without a second factor is only valid internally.

> **Behind a reverse proxy**, the visitor’s real address (`X-Forwarded-For`) counts. In that case, publish the AssetLotse port only to the proxy (e.g. `ASSETLOTSE_BIND=127.0.0.1` for a proxy directly on the server, a shared Docker network for a proxy container) or enter the proxy’s address in `ASSETLOTSE_TRUSTED_PROXIES` – otherwise a computer on the LAN could pretend to be “internal”.

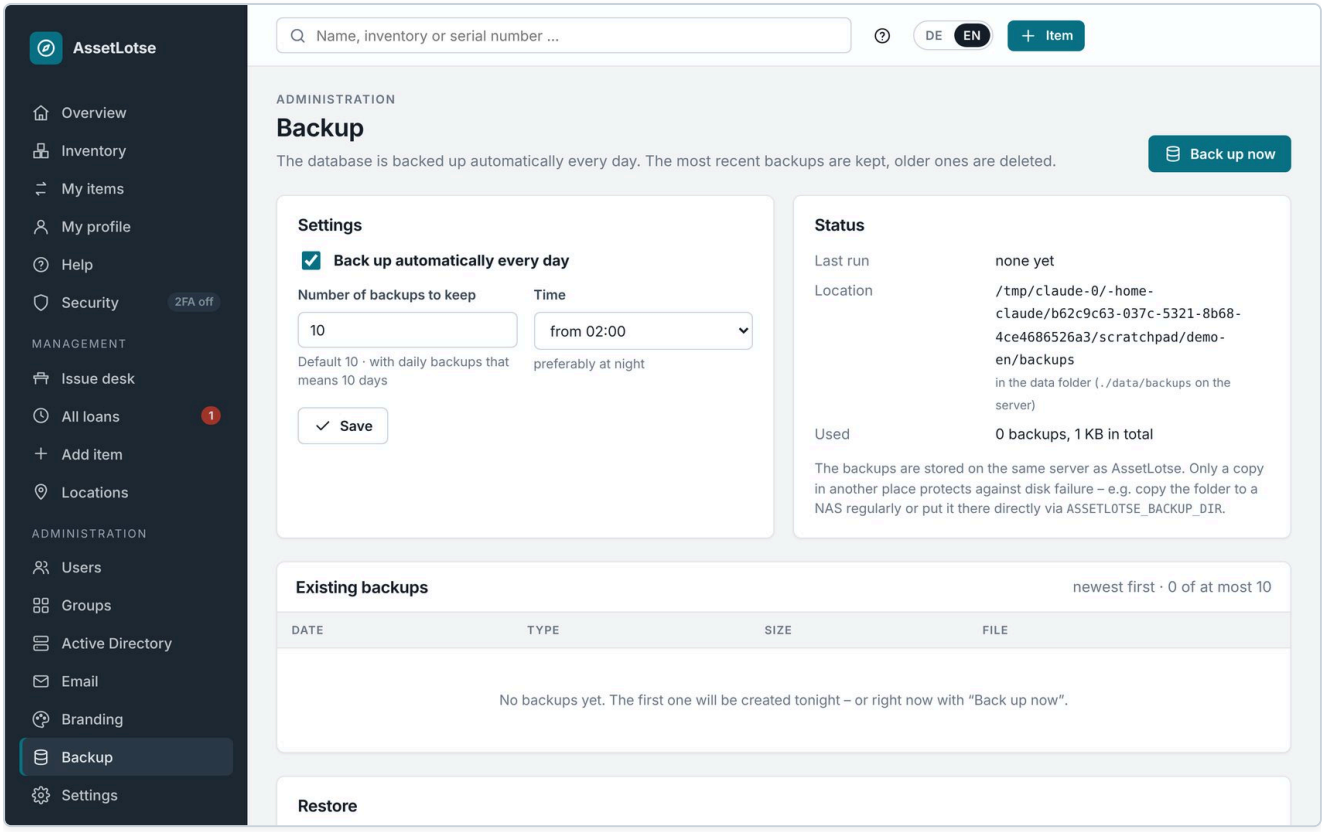
22 Branding



Logo, background image, colors and texts with preview.

Under **Branding** you adapt AssetLotse to your corporate design: **Logo** (PNG with a transparent background or SVG; it also appears in emails and as the browser icon), **Background image** of the sign-in page, **Colours** (main color, accent, sidebar – text colors are chosen automatically for readability) and the **texts** of the sign-in page. Everything is shown immediately in the preview; **Reset everything** restores the original state. The logo and background are publicly accessible – don’t use anything confidential.

23 Backup and restore



Administration → Backup.

The database is **backed up automatically every day** (by default from 2 a.m.) while the system is running, and every backup is checked for corruption. The newest 10 are kept (configurable). **Back up now** creates one immediately – useful before larger changes. If a backup fails, a warning sign appears on the menu item, and administrators receive an email.

- **Location:** `./data/backups`; with `ASSETLOTSE_BACKUP_DIR` e.g. on a mounted NAS drive.
- The database is what gets backed up. Photos, logo and the `secret_key` file are in `./data` – when moving to another server, take the whole folder.
- Backups on the same server protect against operating errors, not against a disk failure: copy the folder to another location regularly.
- The built-in backup is only available for the default **SQLite** database. With MySQL/MariaDB (`DATABASE_URL`), back up with `mysqldump` – and also `./data` with the secret_key.

> **The secret_key file belongs to the database.** It is used to encrypt the BitLocker keys, 2FA secrets and stored passwords. If it is lost, the confidential details are gone for good. Back it up separately and keep it well protected, and always restore a database together with its secret_key. If the key does not match the database, AssetLotse deliberately refuses to start.

Restoring

The current state is backed up automatically first:

```
docker compose stop assetlotse
docker compose run --rm assetlotse flask --app wsgi list-backups
docker compose run --rm assetlotse flask --app wsgi restore-backup FILENAME.db
docker compose start assetlotse
```

24 Security and running on the internet

- **HTTPS:** run AssetLotse behind a reverse proxy with a certificate (e.g. Caddy, Nginx Proxy Manager). AssetLotse detects the proxy automatically. Security keys (WebAuthn) only work over HTTPS. For internet access, only expose the proxy on the router, never the AssetLotse port directly.
- **Sign-in lock:** after 8 failed attempts within 5 minutes, sign-in is locked for 5 minutes; there is also a limit per username across all addresses.
- **AD sessions** are re-checked every 4 hours: anyone who has been disabled in AD or removed from the authorized group is signed out.
- **No third-party servers:** fonts and scripts are part of the program, there is no tracking and only technically necessary cookies.
- **Strict browser protection rules** (Content Security Policy, protection against embedding, CSRF protection); pages for signed-in users are not stored in third-party caches or on the browser's disk.
- **Confidential details** are only shown and changed for accounts with two-factor authentication if the second factor has been confirmed in the current session – even on the internal network.
- **Links in emails** are built only from the configured address (Administration → Email), never from a visitor's request. Without this address, registration with email confirmation is disabled.

25 Privacy

Your privacy policy or record of processing activities should cover what AssetLotse stores:

- name, username, email address and groups of the users,
- loans and returns with dates (who had what and when, who issued it),
- the access log for confidential details and the email log,
- with two-factor authentication, the registered keys,
- with **AD synchronization**, additionally the username, name and email of everyone who would be allowed to sign in – even if they never use AssetLotse (can be turned off on the AD page).

Retention (default 12 months): completed loans, the email log and the access log are deleted automatically after this period; only the number of loans is kept per item. Active and overdue loans are never deleted. Backups contain the data correspondingly longer (with 10 daily backups, up to 10 days). Involve your staff or works council and your data protection officer when introducing the system.

PART D

Help with problems

The most common pitfalls and how to solve them – plus the commands for emergencies.

26 Troubleshooting

The page does not load.

> If `.env` contains `ASSETLOTSE_BIND=127.0.0.1`, AssetLotse is only reachable on the server itself – comment out the line and run `docker compose up -d`. Otherwise, check the port (default 9343, `ASSETLOTSE_PORT`) and the server's firewall (e.g. `sudo ufw allow 9343/tcp`). `docker compose ps` shows whether the container is running; `docker compose logs --tail 50` shows the latest messages.

Someone cannot sign in.

> Under **Users**, check whether the account is active and whether there is a sign-in lock (filter "Sign-in lock (failed attempts)", open the account, then **Lift lock**). For AD accounts, use **Test sign-in** on the AD page: if no rule matches, the person is missing from the authorized group.

Phone with the authenticator app lost.

> Sign in with a recovery code. Without a code: an administrator resets two-factor authentication under **Users → Sign-in and security**.

AD: "The domain controller's certificate is not trusted".

> On the AD page, click **Fetch certificate from DC** and **Trust & save**. If the DC is addressed by IP address, turn off "Verify hostname".

AD: The issue desk cannot find a new colleague.

> New people appear after the next synchronization – or immediately after **Synchronise now** on the AD page. If the cache finds no one, the desk also searches AD live when you submit.

The barcode scanner types wrong characters or does not send Enter.

> Following its manual, set the scanner to the computer's keyboard layout (e.g. US or German) and to "Enter/CR suffix".

The label prints too small or distorted.

> In the print dialog, choose "Actual size" or 100 %, not "Fit to page"; set the label size to 62 × 29 mm.

"Inventory number … is already in use".

> Each number exists only once. The hint below the field links to the item with this number and suggests the next free one.

AssetLotse does not start after an update.

> Read `docker compose logs --tail 50`. If it reports that the key does not match the database, the correct `data/secret_key` is missing. Otherwise, reinstall the previous package with `./update.sh`.

27 Emergency commands

On the server in the program folder, each prefixed with `docker compose exec assetlotse docker-entrypoint.sh` (so they run with the correct permissions):

`flask --app wsgi reset-password admin`	Set a new password and activate the account (also turns an AD account into a local one)
`flask --app wsgi reset-2fa admin`	Reset two-factor authentication
`flask --app wsgi create-admin notfall`	Create an additional local administrator account
`flask --app wsgi backup`	Back up immediately
`flask --app wsgi list-backups`	List backups
`flask --app wsgi send-reminders`	Send reminders immediately
`flask --app wsgi apply-retention`	Apply the retention period immediately

Example: `docker compose exec assetlotse docker-entrypoint.sh flask --app wsgi reset-password admin`

28 Appendix: settings in the .env file

VARIABLE	MEANING
`ASSETLOTSE_PORT`	Port (default 9343, allowed 1024–65535) – applies on the server and in the container.
`ASSETLOTSE_BIND`	Address on which the port is published (default: all). `127.0.0.1` only for a reverse proxy running directly on the same machine (not as a container).
`TZ`	Time zone (default Europe/Berlin).
`PUID / PGID`	User and group for `./data` (default 1000).
`DATABASE_URL`	A different database instead of SQLite, e.g. MySQL/MariaDB – AssetLotse then does not back up by itself (use mysqldump).
`ASSETLOTSE_SECRET_KEY`	Encryption key; if not set, it is generated once and stored in `data/secret_key` (recommended). Never change it afterwards.
`ASSETLOTSE_ACCEPT_NEW_KEY`	`1` = deliberately accept a changed key (one time).
`ASSETLOTSE_TRUSTED_PROXIES`	From whom proxy headers are accepted: `private` (default), `none` or a list of addresses/networks.
`ASSETLOTSE_SECURE_COOKIES`	`auto` (default: with HTTPS), `1` or `0`.
`ASSETLOTSE_BACKUP_DIR`	Folder for the backups, e.g. a NAS drive (mount it as a volume).

VARIABLE	MEANING
ASSETLOTSE_LOGLEVEL	`INFO` (default), `DEBUG` for troubleshooting AD.
ASSETLOTSE_SCHEDULER	`0` = no scheduled jobs in this process (with several app servers).
ASSETLOTSE_SOURCE_URL	Link to the source code on the “About this program” page.

You configure everything else in the user interface.

29 License

AssetLotse is free software under the **GNU Affero General Public License, version 3 or later** (AGPL-3.0-or-later). Anyone may use, share and modify the program. Anyone who distributes a modified version or runs it on a network for others must make its source code available under the same license. AssetLotse is provided without warranty. Version, license and the source code link can be found after signing in under **About this program** (click the version number at the bottom of the sidebar). Bundled font: Inter (SIL Open Font License 1.1).